

① 데이터

X_1	X_2	X_0	y	$w_1 \cdot X_1 + w_2 \cdot X_2 + b$
0	0	1	0	$1 \cdot 0 + 1 \cdot 0 - 1.5 = -1.5$
0	1	1	0	$1 \cdot 0 + 1 \cdot 1 - 1.5 = -0.5$
1	0	1	0	$1 \cdot 1 + 1 \cdot 0 - 1.5 = -0.5$
1	1	1	1	$1 \cdot 1 + 1 \cdot 1 - 1.5 = 0.5$

② 파라미터

- 입력 $X = [0, 0, 1], [0, 1, 1], [1, 0, 1], [1, 1, 1]$
- 정답 $y = [0, 0, 0, 1]$
- 가중치 $W = [w_1, w_2, b] = [0, 0, 0]$ 0으로 시작
- 학습률 $\text{eta} = 0.2$
- 활성화 함수 가중합 > 0 이면 1, 그렇지 않으면 0
- 규칙 $W \leftarrow W + \text{eta} \times \text{error} \times X[i]$
(여기서, $\text{error} = Y[i] - \text{predict}$)

③ Epoch 0, $W = [0, 0, 0]$ I) 입력 $X = [0, 0, 1]$, 정답 $y = 0$

$$\text{합} = 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 1 = 0$$

$$\text{error} = 0 - 0 = 0$$

$$\text{가중치 } W = [0, 0, 0] + 0.2 \cdot 0 \cdot [0, 0, 1] = [0, 0, 0]$$

II) $X = [0, 1, 1]$, $y = 0$

$$\text{합} = 0 \cdot 0 + 1 \cdot 0 + 1 \cdot 0 = 0$$

$$\text{error} = 0 - 0 = 0$$

$$W = [0, 0, 0] + 0.2 \cdot 0 \cdot [0, 1, 1] = [0, 0, 0]$$

III) $X = [1, 0, 1]$, $y = 0$

$$\text{합} = 1 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 = 0$$

$$\text{error} = 0 - 0 = 0$$

$$W = [0, 0, 0] + 0.2 \cdot 0 \cdot [1, 0, 1] = [0, 0, 0]$$

IV) $X = [1, 1, 1]$, $y = 1$

$$\text{합} = 1 \cdot 0 + 1 \cdot 0 + 1 \cdot 0 = 0$$

$$\text{error} = 1 - 0 = 1$$

$$W = [0, 0, 0] + 0.2 \cdot 1 \cdot [1, 1, 1] = \underline{[0.2, 0.2, 0.2]}$$

오차가 0이면
가중치 유지단, 오차가 +
이면, 가중치 ↑반대로 오차가 -
이면, 가중치 ↓

④ Epoch 1, $W = [0.2, 0.2, 0.2, 0.2]$

i) $X = [0, 0, 1]$, $y = 0$, $sum = 0 \cdot 0.2 + 0 \cdot 0.2 + 1 \cdot 0.2 = 0.2$

$y' = 1 (sum = 0.2 > 0)$, $error = 0 - 1 = -1$

$W = [0.2, 0.2, 0.2, 0.2] + 0.2 \times (-1) \times [0, 0, 1] = [0.2, 0.2, 0, 0]$

ii) $X = [0, 1, 1]$, $y = 0$, $sum = 0 \cdot 0.2 + 1 \cdot 0.2 + 1 \cdot 0 = 0.2$

$y' = 1 (sum = 0.2 > 0)$, $error = 0 - 1 = -1$

$W = [0.2, 0.2 + 0, 0] + 0.2 \times (-1) \times [0, 1, 1] = [0.2, 0, -0.2]$

iii) $X = [1, 0, 1]$, $y = 0$, $sum = 1 \cdot 0.2 + 0 \cdot 0 + 1 \cdot (-0.2) = 0$

$y' = 0 (sum = 0)$, $error = 0 - 0 = 0$

$W = [0.2, 0, -0.2] + 0.2 \times 0 \times [1, 0, 1] = [0.2, 0, -0.2]$

iv) $X = [1, 1, 1]$, $y = 1$, $sum = 1 \cdot 0.2 + 1 \cdot 0 + 1 \cdot (-0.2) = 0$

$y' = 0 (sum = 0)$, $error = 1 - 0 = 1$

$W = [0.2, 0, -0.2] + 0.2 \cdot 1 \cdot [1, 1, 1] = [0.4, 0.2, 0]$

⑤ Epoch 2, $W = [0.4, 0.2, 0]$

i) $X = [0, 0, 1]$, $y = 0 \rightarrow sum = 0 \cdot 0.4 + 0 \cdot 0.2 + 1 \cdot 0 = 0$

$y' = 0 (sum = 0)$, $error = 0 - 0 = 0$

$W = [0.4, 0.2, 0] + 0.2 \cdot 0 \cdot [0, 0, 1] = [0.4, 0.2, 0]$

ii) $X = [0, 1, 1]$, $y = 0 \rightarrow sum = 0 \cdot 0.4 + 1 \cdot 0.2 + 1 \cdot 0 = 0.2$

$y' = 1 (sum = 0.2 > 0)$, $error = 0 - 1 = -1$

$W = [0.4, 0.2, 0] + 0.2 \cdot (-1) \cdot [0, 1, 1] = [0.4, 0, -0.2]$

iii) $X = [1, 0, 1]$, $y = 0 \rightarrow sum = 1 \cdot 0.4 + 0 \cdot 0 + 1 \cdot (-0.2) = 0.2$

$y' = 1 (sum = 0.2 > 0)$, $error = 0 - 1 = -1$

$W = [0.4, 0, -0.2] + 0.2 \cdot (-1) \cdot [1, 0, 1] = [0.2, 0, -0.4]$

iv) $X = [1, 1, 1]$, $y = 1 \rightarrow sum = 1 \cdot 0.2 + 1 \cdot 0 + 1 \cdot (-0.4) = -0.2$

$y' = 0 (sum = -0.2 < 0)$, $error = 1 - 0 = 1$

$W = [0.2, 0, -0.4] + 0.2 \cdot 1 \cdot [1, 1, 1] = [0.4, 0.2, -0.2]$

⑥ Epoch 3, $W = [0.4, 0.2, -0.2]$

i) $X = [0, 0, 1]$, $y = 0$, $sum = 0 \cdot 0.4 + 0 \cdot 0.2 + 1 \cdot (-0.2) = -0.2$

$y' = 0$ ($sum = -0.2 < 0$), $error = 0 - 0 = 0$

$W = [0.4, 0.2, -0.2]$ (error=0이므로 W 는 ~~변경~~)

ii) $X = [0, 1, 1]$, $y = 0$, $sum = 0 \cdot 0.4 + 1 \cdot 0.2 + 1 \cdot (-0.2) = 0$

$y' = 0$ ($sum = 0$), $error = 0 - 0 = 0$

$W = [0.4, 0.2, -0.2]$ (error=0)

iii) $X = [1, 0, 1]$, $y = 0$, $sum = 1 \cdot 0.4 + 0 \cdot 0.2 + 1 \cdot (-0.2) = 0.2$

$y' = 1$ ($sum = 0.2 > 0$), $error = 0 - 1 = -1$

$W = [0.4, 0.2, -0.2] + 0.2 \times (-1) \times [1, 0, 1]$
 $= [0.2, 0.2, -0.4]$

iv) $X = [1, 1, 1]$, $y = 1$, $sum = 1 \cdot 0.2 + 1 \cdot 0.2 + 1 \cdot (-0.4) = 0$

$y' = 0$ ($sum = 0$), $error = 1 - 0 = 1$

$W = [0.2, 0.2, -0.4] + 0.2 \times 1 \times [1, 1, 1] = [0.4, 0.4, -0.2]$

⑦ Epoch 4, $W = [0.4, 0.4, -0.2]$

i) $X = [0, 0, 1]$, $y = 0 \rightarrow sum = 0 \cdot 0.4 + 0 \cdot 0.4 + 1 \cdot (-0.2) = -0.2$

$y' = 0$ ($sum = -0.2 < 0$), $error = 0$

$W = [0.4, 0.4, -0.2]$ (error=0이므로 W 는 ~~변경~~)

ii) $X = [0, 1, 1]$, $y = 0 \rightarrow sum = 0 \cdot 0.4 + 1 \cdot 0.4 + 1 \cdot (-0.2) = 0.2$

$y' = 1$ ($sum = 0.2 > 0$), $error = 0 - 1 = -1$

$W = [0.4, 0.4, -0.2] + 0.2 \cdot (-1) \cdot [0, 1, 1] = [0.4, 0.2, -0.4]$

iii) $X = [1, 0, 1]$, $y = 0 \rightarrow sum = 1 \cdot 0.4 + 0 \cdot 0.2 + 1 \cdot (-0.4) = 0$

$y' = 0$ ($sum = 0$), $error = 0 - 0 = 0$

$W = [0.4, 0.2, -0.4]$ (error=0)

iv) $X = [1, 1, 1]$, $y = 0 \rightarrow sum = 1 \cdot 0.4 + 1 \cdot 0.2 + 1 \cdot (-0.4) = 0.2$

$y' = 1$ ($sum = 0.2 > 0$), $error = 1 - 1 = 0$ (정답)

$W = [0.4, 0.2, -0.4]$ (error=0이므로 W 는 ~~변경~~)

subject

딥러닝의 기초

title

3주차 과제. ANN개요에 따른 학습

check ☐ ☐ ☐ ☐ ☐

date 2026. 09. 20

④ Epoch 5, $W = [0.4, 0.2, -0.4]$ 학습 시작.

$$I) X = [0, 0, 1], y = 0 \rightarrow \text{sum} = 0 \cdot 0.4 + 0 \cdot 0.2 + 1 \cdot (-0.4) = -0.4$$

$$y' = 0 (\text{sum} = -0.4 < 0)$$

$$II) X = [0, 1, 1], y = 0 \rightarrow \text{sum} = 0 \cdot 0.4 + 1 \cdot 0.2 + 1 \cdot (-0.4) = -0.2$$

$$y' = 0 (\text{sum} = -0.2 < 0)$$

$$III) X = [1, 0, 1], y = 0 \rightarrow \text{sum} = 1 \cdot 0.4 + 0 \cdot 0.2 + 1 \cdot (-0.4) = 0$$

$$y' = 0 (\text{sum} = 0)$$

$$IV) X = [1, 1, 1], y = 1 \rightarrow \text{sum} = 1 \cdot 0.4 + 1 \cdot 0.2 + 1 \cdot (-0.4) = 0.2$$

$$y' = 1 (\text{sum} = 0.2 > 0)$$

→ 입력 X 에 대한 출력 y' 가 정답 y 와 모두 일치. (오차=0)
가중치 변화 없으며, 수렴 완료

⑤ 결론

$$w_1 = 0.4, w_2 = 0.2, b = -0.4 \text{ (정답)}$$